

FINANCIAL MARKETS & ANALYTICS SUMMARY

Financial markets and quantitative research professional experienced in U.S. fixed income, STIR and options strategies, SOFR/Fed Funds futures, yield curves, repo and liquidity analysis, central-bank policy, and live portfolio risk/P&L management on US\$100K. Skilled in financial modelling, DCF valuation, credit and bond analysis, YTM, backtesting, stress testing, scenario analysis, macroeconomic forecasting and risk reporting using Python, SQL, Excel, Bloomberg Terminal and Power BI. NISM-qualified Investment Adviser and Research Analyst and AMFI-registered Mutual Fund Distributor, with systematic alpha research and a live USD/INR ARIMAX macro-intelligence dashboard integrating market and economic data, model diagnostics and scenario forecasting.

WORK EXPERIENCE

Financial Market Analyst | [Futures First](#) | [LOR](#) Jan'25-Present

- Own SR3/SR1 and SR3/ZQ strategies with full market-risk and P&L responsibility for a US\$100K portfolio, generating US\$50K+ gross P&L through SOFR/Fed Funds basis, curve, butterfly and options-hedging strategies.
- Built and deployed 8+ settlement-accurate backtests using Python, Excel and Bloomberg, incorporating transaction costs, slippage, walk-forward validation, stress testing, liquidity, compounding, roll and convexity.
- Analysed FOMC, CPI, NFP, repo liquidity, Treasury swap spreads and G-SIB funding scenarios, automated P&L, margin, roll-cost and risk reporting, ranked top 20% of 100+ interns, earned a PPO, LOR and performance bonus.

Financial Research Consultant: Innovation & Startup Ecosystems | NITI Aayog, GOI | [LOR](#) Nov'25-Jan'26

- Evaluated startups and 15+ funding and innovation frameworks through business-model, market, unit-economics, cash-flow, liquidity and financial-statement analysis, built DCF valuation and scenario models using forecast free cash flows, WACC and terminal value to assess intrinsic value, funding readiness and investment viability.
- Validated 500+ ANIC records and analysed India's US\$140B+ startup ecosystem, venture flows, sector trends and macroeconomic conditions, delivered 3+ investment and policy reports supporting adequate capital-allocation.

Quantitative Researcher | WorldQuant BRAIN Nov'23-May'24

- Researched 300+ equity and futures alpha hypotheses across price-volume, fundamental and alternative datasets; developed 35+ validated signals using feature engineering, parameter tuning, neutralization, decay and universe filters with transaction-cost, regime, stability and out-of-sample testing.
- Evaluated strategies using Sharpe ratio, information ratio, drawdown, turnover and signal decay, improving hit rate by 13% and response latency by 20%, ranked 56th globally, 8th in India and 1st at NSUT on WQ Leaderboard.

PROJECTS

Rupicast - USD/INR Macro Intelligence & Risk Analytics Dashboard | [Deployed Project Link](#) Jun'26

- Built a Python-Streamlit FX analytics platform processing 10 years of daily USD/INR data into 120+ monthly observations, integrating 7 macroeconomic drivers like U.S./India inflation, Fed and RBI policy rates, crude oil, trade balance and FX reserves through Yahoo Finance and FRED APIs.
- Developed monthly ARIMAX/SARIMAX time-series forecasts with 3-12-month horizons, 95% confidence intervals and 12-24-month rolling backtests, evaluated one-step-ahead predictions using MAE, RMSE and level MAE, alongside ADF stationarity, seasonality and bull/base/bear stress scenarios.

AlphaLens - Market-Neutral Long-Short Factor Strategy Engine | [Deployed Project Link](#) Jun'26

- Built and deployed a Python-Streamlit quantitative backtesting engine on S&P 500 data (2017-2024), combining Value, 12-1 Momentum, VPT Quality and Low-Volatility factors with monthly rebalancing and transaction-cost modelling; delivered 1.125 NAV vs 0.947 benchmark, 3.32% annualised alpha and 0.327 Sharpe, while Kelly sizing achieved 0.415 Sharpe and -7.97% maximum drawdown.
- Engineered $\pm 3\sigma$ signal clipping, RSI regime neutralisation, 10% EWMA volatility targeting, Kelly position sizing and crisis stress testing, reducing drawdown by 4.34 percentage points; generated 19.8% COVID-crash alpha, 18.3% recovery alpha & 1.46 crisis Sharpe, controlling 44.8% monthly turnover and 2.78% transaction-cost drag.

CrediScope AI - Credit Risk Analytics & Loan Default Prediction | [Deployed Project Link](#) May'26

- Analyzed 32,581 borrower records across 11 predictive features, developed and benchmarked 2 machine-learning models Logistic Regression and Random Forest, implemented data imputation, one-hot encoding, feature scaling.

EDUCATION

B.Tech : Netaji Subhas University of Technology, Delhi: Information Technology, 8.95 CGPA, Dept. Rank 1, CVSPK Scholar
Class XII: Kulachi Hansraj Model School, Delhi: CBSE, 95.4%, Scholar Badge recipient for 7 consecutive years

POSITION OF RESPONSIBILITY

Vice-President: Nakshatra | The Astronomical Society of NSUT Jul'23-Jul'24

- Served as the primary organizer for the Triathlon event, coordinating 150+ participants, generating over 80K social impressions, securing 10+ sponsors (GAIL & Boat), contributing to a record-breaking 1,000+ registrations for Resonanz

EXTRA-CURRICULAR ACTIVITIES & ACHIEVEMENTS

- Achieved 2nd place and seed funding by pitching the fintech startup "PayLYT" at NSUT Incubation Centre.